Classwork 2-1: Probability and Expectation

ECON 300 · Applied Statistics for Economists · Stock & Watson Ch. 2

YOUR NAME HERE

2026-09-22

Assigned Thursday, September 24 · **Due** Monday, September 28, 11:59 PM · **Points** 40

Instructions. Do Questions 1–3 *by hand*, showing the arithmetic. Writing out the sum before you evaluate it is the point of the exercise, not the final number. Write your work in math mode inside this document, like this:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i = \frac{3+5+2+6}{4} = \frac{16}{4} = 4$$

Then reproduce every result in R in Question 4. Round to four decimal places.

If your hand answer and your R answer disagree, say so and find the arithmetic error. Reporting a disagreement earns full credit; quietly deleting the hand version earns none.

Submit both `classwork-2-1.qmd` and the rendered `classwork-2-1.pdf` to your course repository and push.

1 Reading the notation

(8 points)

Let $x = (3,\ 5,\ 2,\ 6)$, so $n = 4$.

Compute each of the following by hand. Write out the sum explicitly — as an actual sum of four numbers — before you evaluate it.

(a) $\displaystyle\sum_{i=1}^{4} x_i$, and then $\bar{x} = \frac{1}{n}\sum_{i=1}^{4} x_i$

(b) $\displaystyle\sum_{i=1}^{4} 3x_i$. Compare it to your answer in (a). What rule does this illustrate?

(c) $\displaystyle\sum_{i=1}^{4} x_i^2$

(d) $\displaystyle\left(\sum_{i=1}^{4} x_i\right)^2$

(e) $\sum_{i=1}^{4}(x_i - \bar{x})$

> Parts (c) and (d) are different numbers. Part (e) has a surprise in it — do not round, just compute.

(f) In one sentence: μ and $\bar{x}$ both mean "a mean." What is the difference between them?

2 The distribution of one random variable

(12 points)

A lender's records are organized by census tract. For a randomly drawn application, let X be the number of bank branches in the applicant's tract and let Y equal 1 if the application is denied and 0 if it is approved.

Their **joint probability distribution** is:

	$Y = 0$ (approved)	$Y = 1$ (denied)	$P(X = x)$
$X = 0$	0.10	0.14	
$X = 1$	0.20	0.10	
$X = 2$	0.26	0.06	
$X = 3$	0.12	0.02	
$P(Y = y)$			1.00

(a) Fill in the margins. Sum across each row to get the distribution of X, and down each column to get the distribution of Y. Confirm that each set of margins sums to 1.

(b) Compute the expected number of branches. Write out all four terms.

$$E(X) = \mu_X = \sum_i x_i \, P(X = x_i) = _____$$

(c) Compute $E(X^2)$. Square the **values**; leave the probabilities alone.

$$E(X^2) = \sum_i x_i^2 \, P(X = x_i) = _____$$

(d) Compute the variance and the standard deviation.

$$\text{var}(X) = E(X^2) - [E(X)]^2 = _____ \qquad \sigma_X = \sqrt{\text{var}(X)} = _____$$

(e) Y takes only the values 0 and 1. Using the definition of expected value, show that $E(Y) = P(Y = 1)$, then report $E(Y)$ for this distribution.

3 Two random variables together

(12 points)

Use the same table.

(a) Compute two conditional probabilities:

$$P(Y = 1 \mid X = 0) = \frac{P(X = 0, Y = 1)}{P(X = 0)} = _____ \qquad P(Y = 1 \mid X = 3) = _____$$

(b) In one or two sentences: compare those two numbers to the overall denial rate $P(Y = 1)$ you found in Question 2(e). What does the comparison tell you?

(c) Compute $\text{var}(Y)$ and σ_Y, using the same shortcut as in 2(d). *(For a 0/1 variable, $Y^2 = Y$, so $E(Y^2) = E(Y)$.)*

(d) Compute $E(XY) = \sum_x \sum_y x\, y\, P(X = x, Y = y)$.

Hint: every term with $y = 0$ contributes zero, and so does the $x = 0$ row. Three terms survive.

(e) Compute the covariance and the correlation.

$$\text{cov}(X, Y) = E(XY) - \mu_X \mu_Y = _____ \qquad \rho_{XY} = \frac{\text{cov}(X, Y)}{\sigma_X\, \sigma_Y} = _____$$

(f) Interpret the **sign** of ρ_{XY} in one sentence. Then, in one more sentence, say what this number does **not** establish.

4 Do it all in R

(8 points)

Reproduce every result above in R. Delete `#| eval: false` from each chunk as you complete it, fill in the blanks, and make sure the document renders from top to bottom.

```r
# Question 1
x <- c(3, 5, 2, 6)

n <- length(x)
xbar <- sum(x) / n

sum(x) # 1(a)
sum(3 * x) # 1(b)
sum(x^2) # 1(c)  square first, then add
sum(x)^2 # 1(d)  add first, then square
sum(x - xbar) # 1(e)
```

```r
# Question 2 - the joint distribution as a matrix.
# Rows are X = 0,1,2,3.  Columns are Y = 0,1.
J <- matrix(
  c(0.10, 0.14, 0.20, 0.10, 0.26, 0.06, 0.12, 0.02),
  nrow = 4,
  byrow = TRUE,
  dimnames = list(X = 0:3, Y = 0:1)
```

```
)

xvals <- 0:3
yvals <- 0:1

sum(J) # must be 1

px <- ____ # the margins of X   (hint: rowSums)
py <- ____ # the margins of Y   (hint: colSums)

px
sum(px)
py
sum(py)
```

```
# Question 2(b)-(e)
EX <- sum(____ * px) # E(X)
EX2 <- sum(____ * px) # E(X^2)

varX <- EX2 - ____
sdX <- sqrt(varX)

EY <- sum(yvals * py)

c(EX = EX, EX2 = EX2, varX = varX, sdX = sdX, EY = EY)
```

```
# Question 3
J["0", "1"] / px["0"] # 3(a)  P(Y = 1 | X = 0)
J["3", "1"] / ____ #       P(Y = 1 | X = 3)

varY <- EY * (1 - EY) # 3(c)
sdY <- sqrt(varY)

EXY <- sum(outer(xvals, yvals) * J) # 3(d)  outer() builds the table of x*y

covXY <- EXY - ____ * ____ # 3(e)
rho <- covXY / (sdX * ____)

c(varY = varY, sdY = sdY, EXY = EXY, cov = covXY, rho = rho)
```

Below your final chunk, write two or three sentences comparing your hand answers to your R answers. Name any disagreement and say what caused it.

Submission checklist

- ☐ Every `#| eval: false` removed and every blank filled in
- ☐ Document renders top to bottom with no errors
- ☐ Every hand calculation written in math mode, with the sum written out

☐ Every "in one sentence" question actually answered in prose
☐ Your name in the YAML header
☐ Both `classwork-2-1.qmd` and `classwork-2-1.pdf` committed and pushed